



GUJARAT TECHNOLOGICAL UNIVERSITY

Shree Swami Atmanand Saraswati Institute of Technology

A

Micro Project Report on

Library Book Data Analyzer

Under subject of

Python for Data Science (BE05000231)

B.E 3rd year, semester – V

COMPUTER ENGINEERING

Under the Guidance of

(Prof. Vipulkumar Kania)

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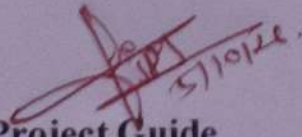
CERTIFICATE

This is to certify that the Project entitled “Library Book Data Analyzer” has been prepared by **Dhokiya Hiral (240760107043)**, **Chudasama Bhavyaa (240760107028)**, **Thummar Archita(240760107006)**, **Goti Khushi (240760107060)**, **Guna Khushi (240760107064)**, **Dhameliya Dhruvi (240760107037)** in the subject of **Python for Data Science (BE05000231)** under my guidance in partial fulfillment of the degree of **Bachelor of Engineering in Computer Engineering Department (5th Semester)** of Gujarat Technological University, Ahmedabad

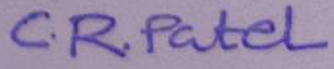
SSASIT

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Place: SSASIT, Surat


Project Guide

Prof. (Vipulkumar Kania)



Head of Department

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ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our faculty member for giving us the opportunity to work on the micro project "Library Book Data Analyzer" as a part of the subject Python for Data Science.

This project helped us understand the practical applications of Python in data analysis and visualization. During the development of this project, we learned how to work with datasets, perform data preprocessing, calculate statistical values, analyze data using Pandas, and create visualizations using Matplotlib.

We are also thankful to our college and department for providing us with the required facilities and guidance to complete this project successfully.

Submitted by:

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ABSTRACT

The Library Book Data Analyzer is a Python-based micro project developed for the subject Python for Data Science. The main purpose of this project is to analyze library book data and convert raw data into useful information through data processing, statistical analysis, visualization, and dashboard development.

Libraries contain information about a large number of books, including book names, authors, genres, publication years, and ratings. When such information is stored in a raw CSV file, it can be difficult to understand the overall characteristics of the collection. Therefore, this project provides a simple system that automatically analyzes the available library data and presents the results graphically.

The project uses Python, Pandas, NumPy, and Matplotlib. Pandas is used for loading and processing the CSV dataset, NumPy is used for numerical operations, and Matplotlib is used to create different charts and visualizations.

The dataset contains information such as Book ID, Book Name, Author, Genre, Publication Year, and Rating. The system calculates important information such as total number of books, average rating, number of genres, number of authors, genre-wise book count, publication-year distribution, rating distribution, and author-wise book count.

The analyzed information is presented through a Library Book Data Analysis Dashboard. The dashboard contains KPI cards and graphical charts that provide a quick overview of the dataset. This makes the information easier to understand compared to analyzing the raw CSV file directly.

The project demonstrates the practical application of Python programming and data science concepts for solving a simple real-world data analysis problem.

PROBLEM STATEMENT

Libraries contain a large amount of information about books, such as book name, author, genre, publication year, rating, and book identification number. When the number of books increases, managing and understanding this information manually becomes difficult and time-consuming. A raw dataset may contain hundreds of records, but simply storing the data does not provide useful information about the overall library collection.

Traditional manual analysis requires checking individual records and performing calculations separately. This makes it difficult to quickly answer questions such as how many books are available, which genre contains the most books, what the average rating is, which authors have the highest number of books, and how books are distributed across different publication years. It may also be difficult to identify patterns and trends by looking at raw tabular data.

Another problem is that the important information is not available in a single visual representation. Users may need to perform separate calculations to understand different aspects of the dataset. This can make the analysis less efficient and difficult for users who are not familiar with handling raw data.

Therefore, there is a need for a simple and efficient Library Book Data Analyzer that can automatically process the library dataset and convert raw data into meaningful information. The system should be able to load the dataset, check and preprocess the data, perform statistical calculations, analyze different book attributes, and generate graphical visualizations.

The proposed system uses Python, Pandas, NumPy, and Matplotlib to solve these problems. It analyzes the dataset based on genre, rating, publication year, and author and calculates important statistical values such as total books, average rating, total genres, and total authors.

To make the analyzed information easier to understand, the system also provides a dashboard containing KPI cards, charts, and key insights. The dashboard gives a summarized view of the library dataset and helps users understand important patterns without manually examining every record.

Thus, the main problem addressed by this project is to transform raw library book data into useful, organized, and visually understandable information through Python-based data analysis and dashboard visualization.

OBJECTIVES

The main objectives of the project are:

1. To collect and organize library book data

The project uses a structured CSV dataset containing information about books, authors, genres, publication years, and ratings.

2. To load the dataset using Python

The project uses Pandas to read the CSV file and store the data in a DataFrame.

3. To perform data preprocessing

The dataset is checked for missing values, incorrect data, duplicate records, and appropriate data types before performing analysis.

4. To perform statistical analysis

The project calculates important statistics such as:

- Total books
- Average rating
- Total genres
- Total authors

5. To analyze different characteristics of books

The system analyzes books based on:

- Genre
- Publication year
- Author
- Rating

6. To visualize the analyzed data

Charts are created using Matplotlib so that patterns and trends can be easily understood.

7. To develop a dashboard

A dashboard is created that combines important statistics and charts in one place.

8. To identify useful insights

The project identifies patterns such as the most common genres, publication trends, rating distribution, and authors with more books.

9. To gain practical knowledge

The project helps students understand the practical use of:

- Python
- Pandas
- NumPy
- Matplotlib
- Data analysis
- Data visualization
- Dashboard development

LITERATURE SURVEY

Data analysis is the process of examining, processing, and interpreting data to obtain useful information. With the increasing amount of digital data, data analysis has become important in areas such as education, business, healthcare, libraries, banking, and many other fields.

Python is one of the commonly used programming languages for data analysis because it provides several libraries specifically designed for handling and analyzing data.

Pandas

Pandas is a Python library used for handling structured data. It provides a powerful data structure called DataFrame, which is suitable for working with tabular data.

In this project, Pandas is used to:

- Read the CSV file.
- Store the dataset.
- Check data.
- Filter records.
- Count values.
- Group data.
- Calculate statistics.

NumPy

NumPy is a numerical computing library for Python. It provides support for numerical calculations and array-based operations.

It can be used for:

- Mathematical calculations.
- Numerical processing.
- Statistical operations.
- Working with numerical arrays.

Matplotlib

Matplotlib is a Python library used for data visualization. It allows numerical information to be represented through graphical charts.

In this project, Matplotlib is used to create:

- Bar charts
- Horizontal bar charts
- Line charts
- Rating distribution charts

Data Visualization

Data visualization helps users understand large amounts of information quickly. Instead of examining hundreds of rows manually, charts can show patterns and comparisons clearly.

Dashboard

A dashboard combines important statistics and visualizations into one interface. In this project, the dashboard provides a quick summary of the library dataset using KPI cards and charts.

Thus, the proposed project combines the concepts of data processing, statistical analysis, visualization, and dashboard development into a single application.

DATASET DESCRIPTION

The dataset used in the Library Book Data Analyzer project is a structured library book dataset stored in CSV (Comma-Separated Values) format. The dataset is generated using Python and is used as the primary input for performing data analysis, statistical calculations, data visualization, and dashboard generation.

The dataset contains 150 book records. Each record represents one book and contains different attributes that describe the book, such as its name, author, genre, publication year, rating, available copies, price, and language. The generated dataset is stored in the file `library_books.csv` inside the data folder.

5.1 Dataset Size

The dataset consists of:

- Total number of books/records: 150
- Total number of attributes: 9
- File format: CSV
- Dataset file: `library_books.csv`

The dataset generation program creates records from 1 to 150, resulting in a total of 150 book records.

The dataset contains the following 9 attributes:

- Book_ID
- Book_Name
- Author
- Genre
- Publication_Year
- Rating
- Available_Copies
- Price
- Language

These attributes are defined in the *Pandas Data Frame* used to create the dataset.

5.2 Description of Dataset Attributes

1. Book_ID

Book_ID is the unique identification number assigned to each book in the dataset. It helps distinguish one book record from another.

The IDs are generated in the format:

B001
B002
B003
...
B150

The project uses a formatted ID for every generated book record.

2. Book_Name

Book_Name represents the title or name of the book.

The dataset generator contains a collection of different book titles related to areas such as programming, technology, science, fiction, self-help, business, and other subjects.

Some examples included in the dataset generator are:

- Python Basics
- Data Science Handbook
- Introduction to AI
- Machine Learning Essentials
- The Alchemist
- Atomic Habits
- Clean Code
- Deep Work
- Digital Fortress
- The Great Gatsby
- Wings of Fire
- Rich Dad Poor Dad
- The Power of Habit
- Artificial Intelligence

- Computer Networks
- Database Systems
- Web Development Guide
- Statistics for Data Science
- Introduction to Algorithms
- The Psychology of Money

This attribute can be used to identify and display the names of individual books.

3. Author

Author represents the author associated with each book.

The dataset generator contains 15 possible author names. These include:

- John Smith
- Robert Brown
- Emily Johnson
- David Wilson
- Sarah Davis
- Michael Miller
- James Anderson
- Linda Thomas
- William Taylor
- Patricia Moore
- Daniel Martin
- Jennifer Lee
- Christopher White
- Jessica Harris
- Matthew Clark

The author attribute can be used to perform author-wise analysis, such as finding the number of books associated with each author and identifying the top authors in the dataset.

4. Genre

Genre represents the category or type of a book.

The dataset contains 8 different genres:

- Programming
- Technology

- Science
- Fiction
- History
- Biography
- Self-Help
- Business

Genre is an important attribute for analysis because it allows the project to determine how the 150 books are distributed among different categories.

For example, the dashboard can use a Books by Genre chart to visually compare the number of books belonging to each genre.

5. Publication_Year

Publication_Year represents the year in which a book was published.

The dataset generates publication years between 2010 and 2025.

This attribute can be used to analyze the distribution of books over different years. A line chart can be created to show how the number of books changes across publication years.

6. Rating

Rating represents the rating assigned to each book.

The dataset generator creates ratings between 3.0 and 5.0, with values rounded to one decimal place.

The rating attribute is useful for:

- Calculating the average rating.
- Studying rating distribution.
- Comparing books based on ratings.
- Creating a rating visualization in the dashboard.

7. Available_Copies

Available_Copies represents the number of copies of a particular book available in the dataset.

The generated value ranges from 1 to 15 copies.

This attribute can be useful for analyzing book availability. It can also be used in future

improvements of the project to identify books with low or high availability.

8. Price

Price represents the price associated with each book.

The dataset generator generates book prices between ₹200 and ₹1000.

The price attribute can be used for additional analysis, such as finding the average price or comparing the prices of different books.

9. Language

Language represents the language of a book.

The dataset contains three possible languages:

- English
- Hindi
- Gujarati

This attribute can be used to analyze how the books are distributed according to language.

SYSTEM DESIGN

6.1 System Overview

The Library Book Data Analyzer is a Python-based data analysis and visualization system developed for analyzing library book data. The system takes a structured CSV dataset as input and processes the information using Python libraries such as Pandas, NumPy, and Matplotlib.

The dataset contains 150 library book records with information such as Book ID, Book Name, Author, Genre, Publication Year, Rating, Available Copies, Price, and Language. The dataset consists of 9 attributes.

The main purpose of the system is to convert raw library data into meaningful statistical information and graphical representations. Instead of manually examining every book record, the system automatically performs data processing and generates useful information such as:

- Total number of books
- Average book rating
- Total number of genres
- Total number of authors
- Number of books in each genre
- Distribution of book ratings
- Number of books published in each year
- Top authors based on number of books
- Important insights from the dataset

The final output is presented through a Library Book Data Analysis Dashboard, which provides a summarized and visual representation of the dataset.

6.2 System Architecture

The system follows a simple layered architecture in which the input dataset passes through different processing stages before the final dashboard is generated.

Library Book Dataset
Library_books.csv



Data loading
Pandas



Data Cleaning and Preprocessing
Data Validation and Checking



Data Analysis
Statistics and Grouping



Data Visualisation
Matplotlib



Dashboard generation
KPI's + charts + insights



Final result
Analysis and interpretation

The system loads the CSV dataset, checks and cleans the data, performs statistical analysis, generates visualizations, and presents the results in a dashboard.

6.3 System Modules

The system consists of seven main modules:

1. Dataset Loading

The library dataset is loaded from a CSV file using the Pandas library. The CSV file library_books.csv contains information about books such as Book ID, Book Name, Author, Genre, Publication Year, Rating, Available Copies, Price, and Language. The read_csv() function is used to load the dataset into a Pandas DataFrame for further processing and analysis.

```

20 # LOAD DATASET
21 #
22
23 try:
24     df = pd.read_csv('data/library_books.csv')
25
26     print("\n")
27     print("Library books.csv file loaded.")
28     print("\nDataset loaded successfully.")
29     print("Total records:", len(df))
30
31 except FileNotFoundError:
32     print("\nError: Library books.csv file not found.")
33     print("Please check the 'data' folder and file name.")
34
35 except Exception as e:
36     print("\nError while loading dataset:", e)
37
38 finally:
39     print("Dataset loading process completed.")
40
41
42

```

figure 1:dataset loading

2. Data Validation and Error Handling

The system validates user inputs such as Book ID, rating, price, and available copies. It also checks duplicate Book IDs and uses try-except-finally blocks to handle invalid inputs and errors safely.

```

663     try:
664         rating = float(
665             input("\nEnter minimum rating: ")
666         )
667
668         if rating < 0 or rating > 5:
669             print("Rating must be between 0 and 5.")
670             return
671
672         result = df[
673             df["Rating"] >= rating
674         ].sort_values(
675             by="Rating",
676             ascending=False
677         )
678
679         if result.empty:
680             print("\nNo books found.")
681
682         else:
683             print(
684                 "\nBooks with rating >= ",
685                 rating
686             )
687
688             print(
689                 result.to_string(index=False)
690             )
691
692             print("\ntotal books:", len(result))
693
694     except ValueError:
695         print("\nPlease enter a valid number.")
696

```

Figure 2:Error handling code

3.Data Analysis

The system calculates:

- Total books
- Total authors and genres
- Average rating
- Total available copies
- Average book price
- Books by genre
- Top 10 authors
- Publication-year distribution
- Highest/lowest rated books
- Most expensive/cheapest books

4.Advanced Analysis

Pandas and NumPy are used to calculate:

- Average rating by genre
- Available copies by genre
- Minimum, maximum, and average ratings
- Minimum, maximum, and average prices

5.Data Visualization

The system generates six visualizations:

- Number of Books by Genre
- Books Published by Year
- Rating Distribution
- Top 10 Authors
- Average Rating by Genre
- Available Copies by Genre

6.Dashboard Generation

Important results are combined into a single dashboard containing KPI values, charts, and key insights.

Dashboard Design

The final dashboard contains four KPI cards:

- Total Books – 150
- Average Rating – 3.90
- Total Genres – 8
- Total Authors – 15

It also includes:

- Books by Genre – shows genre-wise book distribution.
- Rating Distribution – shows the distribution of book ratings.
- Books by Publication Year – shows the number of books published in different years.
- Top 10 Authors – shows authors with the highest number of books.
- Key Insights – summarizes important findings.

7. User Interaction and CRUD Operations

The system allows users to add, search, update, and delete book records through a simple menu-driven interface. Users can enter new book details, search existing records, modify information, or remove a book from the dataset. The system also validates user input and saves changes to the CSV file

```
PS C:\Users\Viraj\OneDrive\Desktop\PDS\data> python library_book_analyzer.py

LIBRARY BOOK DATA ANALYZER

Dataset loaded successfully.
Total records: 151
Dataset loading process completed.

LIBRARY BOOK DATA ANALYZER

1. Display All Books
2. Add New Book
3. Search Book by ID
4. Search Book by Name
5. Display Books by Author
6. Display Books by Genre
7. Display Books by Language
8. Update Book
9. Delete Book
10. Library Statistics
11. Rating and Price Analysis
12. Advanced Analysis
13. Namey Analysis
14. Filter Books by Rating
15. Sort Books
16. Generate Visualizations
17. Generate Dashboard
18. Exit

Enter your choice: 2
```

Figure 3: Library Book Data Analyzer all operations

17. Generate Dashboard
18. Exit

Enter your choice: 2

ADD NEW BOOK

Enter Book ID: B151
Enter Book Name: Python for data science
Enter Author Name: David
Enter Genre: science
Enter Publication Year: 2023
Enter Rating (0-5): 4.2
Enter Available Copies: 4
Enter Price: 699
Enter Language: english

Book added successfully!

New Book Details:

Book_ID	Book_Name	Author	Genre	Publication_Year	Rating	Available_Copies	Price	Language
B151	Python for data science	David	science	2023	4.2	4	699.0	english

Add book operation completed.

Figure 4: Add new book in library

6.4 Dashboard Implementation

The dashboard is generated using Python visualization libraries and combines the analyzed data into a single graphical interface.

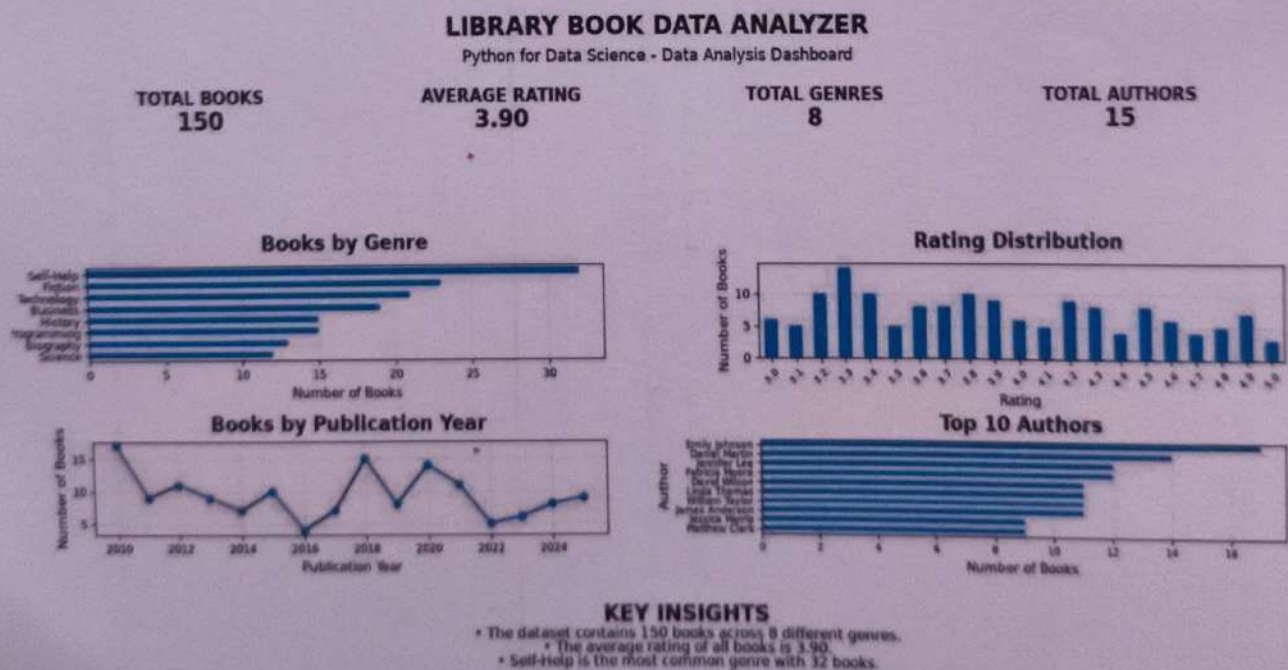


Figure 6.4: Library Book Data Analysis Dashboard

The dashboard provides a quick and clear understanding of the library dataset through numerical values, charts, and summarized insights.

6.5 Advantages

- Reduces manual data analysis.
- Performs calculations automatically.
- Provides clear graphical representation.
- Makes large datasets easier to understand.
- Combines important results in one dashboard.
- Can be extended with additional analysis.

RESULT

The Library Book Data Analyzer successfully processed and analyzed the library book dataset using Python.

7.1 User Interaction

The system provides a menu-driven interface that allows the user to select different operations such as adding, searching, updating, deleting, filtering, and analyzing book records.

```
=====
LIBRARY BOOK DATA ANALYZER
=====

1. Display All Books
2. Add New Book
3. Search Book by ID
4. Search Book by Name
5. Display Books by Author
6. Display Books by Genre
7. Display Books by Language
8. Update Book
9. Delete Book
10. Library Statistics
11. Rating and Price Analysis
12. Advanced Analysis
13. NumPy Analysis
14. Filter Books by Rating
15. Sort Books
16. Generate Visualizations
17. Generate Dashboard
18. Exit

Enter your choice: 2
```

Figure 7.1: Main Menu of Library Book Data Analyzer

7.2 Search Operation

The search functionality allows users to find specific book records using details such as Book ID, book name, or author. The system displays the matching records based on the user's input.

1. Display All Books
2. Add New Book
3. Search Book by ID
4. Search Book by Name
5. Display Books by Author
6. Display Books by Genre
7. Display Books by Language
8. Update Book
9. Delete Book
10. Library Statistics
11. Rating and Price Analysis
12. Advanced Analysis
13. Numpy Analysis
14. Filter Books by Rating
15. Sort Books
16. Generate Visualizations
17. Generate Dashboard
18. Exit

Enter your choice: 4

Enter Book Name: Python for data science

SEARCH RESULT

Book ID	Book Name	Author	Genre	Publication Year	Rating	Available Copies	Price	Language
1151	Python for data science	David	science	2023	4.2	4	699.0	english

Books found: 1

Book name search completed.

Figure 7.2: Book Search Result

7.3 Library Statistics and Analysis

The system analyzes the library data and displays important statistics such as total books, average rating, genres, and authors.

```
Total Books: 152
Total Authors: 17
Total Genres: 9
Average Rating: 3.9
Total Available Copies: 1118
Average Book Price: 584.63
```

Genre Count:

```
Genre
Self-Help      12
Fiction        23
Technology     21
Business       19
Programming    15
History        15
Biography      11
Science        12
science        2
Name: count, dtype: int64
```

Author Count:

```
Author
Emily Johnson      12
Daniel Martin      14
Patricia Moore      12
Jennifer Lee        12
David Wilson        11
Linda Thomas        11
William Taylor      11
James Anderson      11
Jessica Harris       9
Matthew Clark        9
Robert Brown         8
Christopher White    7
Michael Miller       7
John Smith           6
Sarah Davis          5
david                1
```

Figure 7.3: Library Statistics and Analysis Output

The main results obtained from the analysis are:

- The dataset contains 150 books.
- There are 8 different genres and 15 authors.
- The calculated average rating is 3.90.
- The dashboard shows that Self-Help is the most common genre with 32 books.
- The system analyzed the distribution of books by genre, rating, publication year, and author.
- The program also identified the highest and lowest rated books and the most expensive and cheapest books.
- Additional analysis was performed for average rating by genre and available copies by genre.
- Six different visualizations were generated to represent the analyzed data.
- A final dashboard was created containing KPI values, charts, and key insights.

The dashboard provides a summarized and visual representation of the complete dataset, making the analysis easier to understand.

Conclusion

The Library Book Data Analyzer successfully demonstrates the practical use of Python for data analysis, data processing, and visualization. The project uses libraries such as Pandas, NumPy, Matplotlib, and Seaborn to work with library book data stored in a CSV file.

The system provides different operations for managing and analyzing the dataset, including searching, adding, updating, deleting, filtering, sorting, and analyzing book records. It also performs statistical analysis to obtain useful information such as the total number of books, average rating, number of genres, and number of authors.

The project converts raw library data into meaningful statistical results and graphical representations. The generated visualizations and dashboard make the information easier to understand and help users identify important characteristics of the library collection without manually examining individual records.

Overall, the project provides a simple and practical application of Python for Data Science. It can be further improved by adding features such as more detailed price and language analysis, interactive dashboards, advanced filtering, real-time data updates, and support for larger datasets.